

CPD Plan

Competency 1: A2 - Developing technological solutions to unusual or challenging problems

Justification: My MEng and MSc projects have involved tackling technically challenging problems across robotics, AI, and embedded systems, placing me at Working level. As I move into doctoral research, I will need to develop further towards Practitioner level - independently framing and solving open research problems with no established solutions, under uncertainty and with significant technical risk.

Potential Future Activity: Engage with research literature beyond coursework-level knowledge and attend robotics conferences and workshops to understand the state of the art. Through my PhD, progressively take on open research problems of increasing complexity to build independent research capability.

Competency 2: C3 - Lead teams or technical specialisms and assist others to meet changing technical and managerial needs

Justification: My project experience has involved working in teams and contributing technically, but I have not yet had formal responsibility for leading a team or mentoring others. As I progress into a PhD and research environment, I will increasingly need to take on leadership roles - whether coordinating with collaborators, supervising students, or leading technical work packages.

Potential Future Activity: Take on leadership responsibilities within my PhD research group, such as coordinating collaborative work or mentoring newer students. Seek out opportunities through the CDT to develop leadership and supervision skills, and engage with any relevant training offered by the university.

Competency 3: D2 - Clearly present and discuss proposals, justifications and conclusions

Justification: I have communicated technical work through reports/papers, placing me at Working level. As I transition into research, I will need to present increasingly complex and original contributions to diverse audiences - supervisors, conference attendees, and potential collaborators. Strengthening

this competency is essential for effectively communicating my research.

Potential Future Activity: Present work regularly at internal research group meetings and CDT seminars. Submit to and hopefully attend international conferences. Take advantage of presentation and research communication training offered through the CDT or the university.